

```

"description": "",
"main": "index.js",
"scripts": {
  "test": "echo \"Error: no test specified\" && exit 1"
},
"dependencies": {
  "serialport": "^4.0.0"
},
"author": "",
"license": "ISC"
}

```

Why Yarn?

npm is good, but it also has some shortcomings. Here is a list of some of them.

- **Nested dependencies:** npm version 2 nests dependencies, which leads to duplicates. For users of Windows, file paths may get long, which causes problems with Windows when trying to delete them. To fix this problem, you should use npm version 3.
- **Queued install:** It installs each dependency one after the other, which may take up a lot of time.
- **Single registry:** If a package is not on the npm registry, then forget about downloading it through npm.
- **No offline installation:** Every time you have to download dependencies from the npm registry, a working Internet connection is required, which takes up a lot of time and consumes more bandwidth. Yarn overcomes most of the shortcomings of npm and provides additional power to manage dependencies. Here are the powerful features that Yarn offers.
- **Ultra-fast:** It caches every package it downloads, so it never needs to download it again. It also parallelises operations to maximise resource utilisation; so install times are faster than ever.
- **Offline mode:** If you've installed a package before, you can install it again without any Internet connection.
- **Extremely secure:** Yarn uses *checksums* to verify the integrity of every installed package before its code is executed.
- **Super reliable:** Using a detailed, but concise, lockfile format, and a deterministic algorithm for installs, Yarn is able to guarantee that an install that worked on one system will work exactly the same way on any other system.
- **Same packages:** You can install any package from npm and keep your package workflow the same.
- **Network resilience:** A single request failing will not cause an install to fail. Requests are retried upon failure.
- **Flat mode:** You can resolve mismatched versions of dependencies to a single version to avoid creating duplicates.

Yarn installation

Before using Yarn, you'll first need to install it on your

system. There is an increasing number of ways to install Yarn. But in this article, I will cover only some of the options.

1. **Installation on Windows:** This is done through the installer *.msi* file. Download it using the link <https://yarnpkg.com/latest.msi>
Install via Chocolatey (package manager for Windows):

```
choco install yarn
```

Install via Scoop (command-line installer for Windows):

```
scoop install yarn
```

2. **Installation on Centos, Fedora and RHEL Linux:** Install it via the RPM package repository.
Step 1:

```
sudo wget https://dl.yarnpkg.com/rpm/yarn.repo -O /etc/yum.repos.d/yarn.repo
```

Step 2:

```
sudo yum install yarn
```

3. **Installation on MacOS:** Install Yarn through the Homebrew package manager. This will also install Node.js if it is not already installed.

```
brew install yarn
```

Once Yarn is installed, then by using the command given below, we can check what version it is:

```
yarn --version
```

As of now, Yarn's current stable version is v0.27.5 and the latest unstable version is v.28.4 (Nightly Build – this release may have bugs).

Installing project dependencies

Here is a list of possible Yarn commands that help us to add and install project dependencies.

1. **yarn add:** This command installs a package and any packages that it depends on. It installs the dependencies in the local *node_modules* directory and also an entry in the *package.json* and *yarn.lock* files. So members of a team working on the same project can get the same modules installed on their machines by executing the *yarn* or *yarn install* commands.
2. **yarn install:** This is used to install all dependencies for a project listed within *package.json* in the *node_modules* folder. It is most commonly used when you have just checked out code for a project, or when another developer on the project has added a new dependency that you need to pick up.